

1) The first type has 30% of X, 20% of Y and 50% of Z by weight.
 The second type has 40% of X, 10% of Y and 30% of Z.
 The cost of two scraps are Rs 120 and 160 per kg respectively.

The company requires at least 240 kg of material X, 100 kg of material Y and 290 kg of material Z.

Let's say, x_1 kg of scrap 1 and x_2 kg of scrap 2 is purchased.

$C = 120x_1 + 160x_2 \rightarrow$ Objective is to minimize the cost.
 function

Constraints: Composition yields of each material $\rightarrow$

X: $0.30x_1 + 0.40x_2 \geq 240$

Y: $0.20x_1 + 0.10x_2 \geq 100$

Z: $0.50x_1 + 0.30x_2 \geq 290$

Regular constraints

and $x_1, x_2 \geq 0$ } Non-negativity constraints

$3x_1 + 4x_2 \geq 2400$

$2x_1 + x_2 \geq 1000$

$5x_1 + 3x_2 \geq 2900$

when $x_1 = 0$, $x_2 = 600$

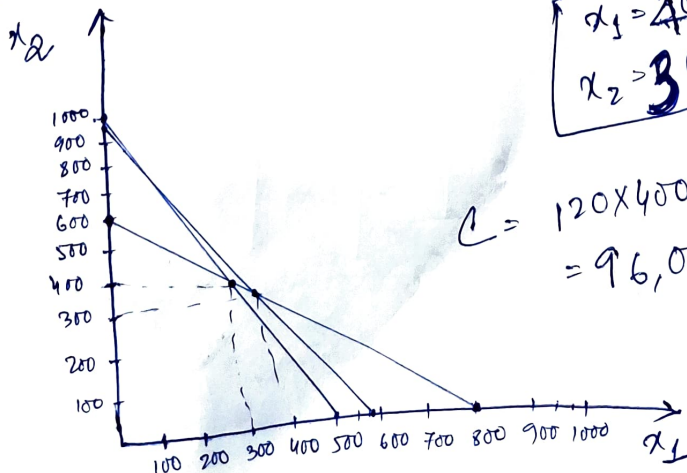
when $x_2 = 0$, $x_1 = 800$

when $x_1 = 0$, $x_2 = 1000$

when $x_2 = 0$, $x_1 = 500$

when $x_1 = 0$, $x_2 = \frac{2900}{3}$

when $x_2 = 0$, $x_1 = 580$



$x_1 = 400$
 $x_2 = 300$

$C = 120 \times 400 + 160 \times 300$
 $= 96,000.$

Minimize

1 b) $z = 3a + 5b$

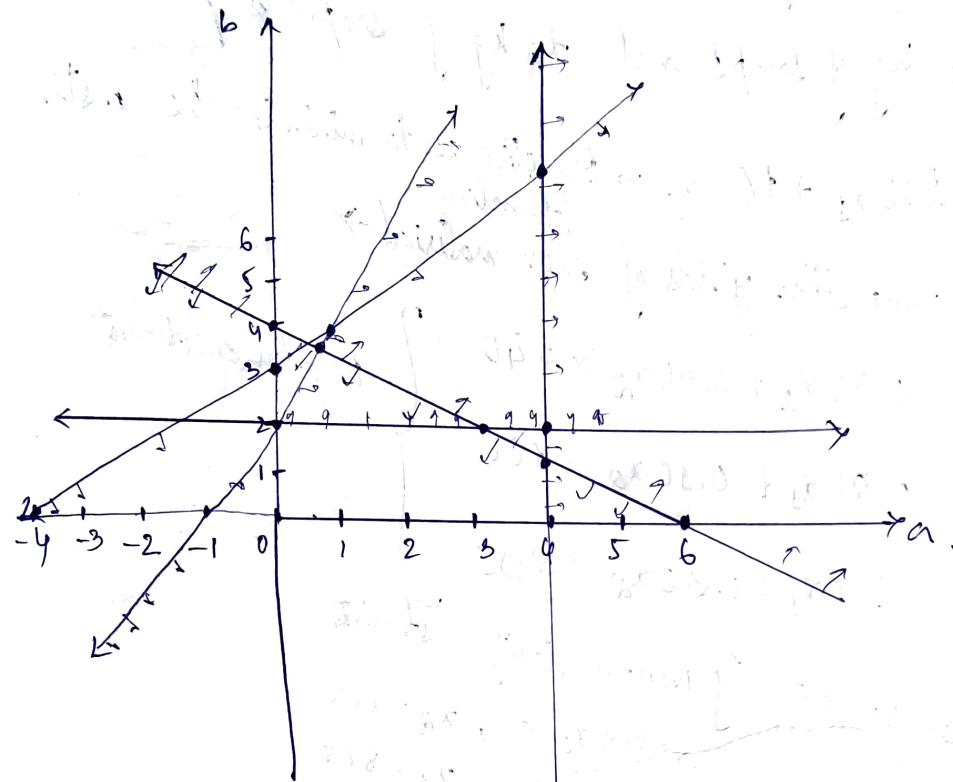
$-3a + 4b \leq 12$ $\begin{cases} a=0, b=3 \\ a=-4, b=0 \end{cases}$

$2a - b \geq 2 \Rightarrow 2a - b \geq -2 \Rightarrow -2a + b \leq 2$ $\begin{cases} a=0, b=2 \\ a=-1, b=0 \end{cases}$

$2a + 3b \geq 12 \Rightarrow -2a - 3b \leq -12$ $\begin{cases} a=0, b=4 \\ a=6, b=0 \end{cases}$

$1a + 0b \geq 4 \Rightarrow a \geq 4$

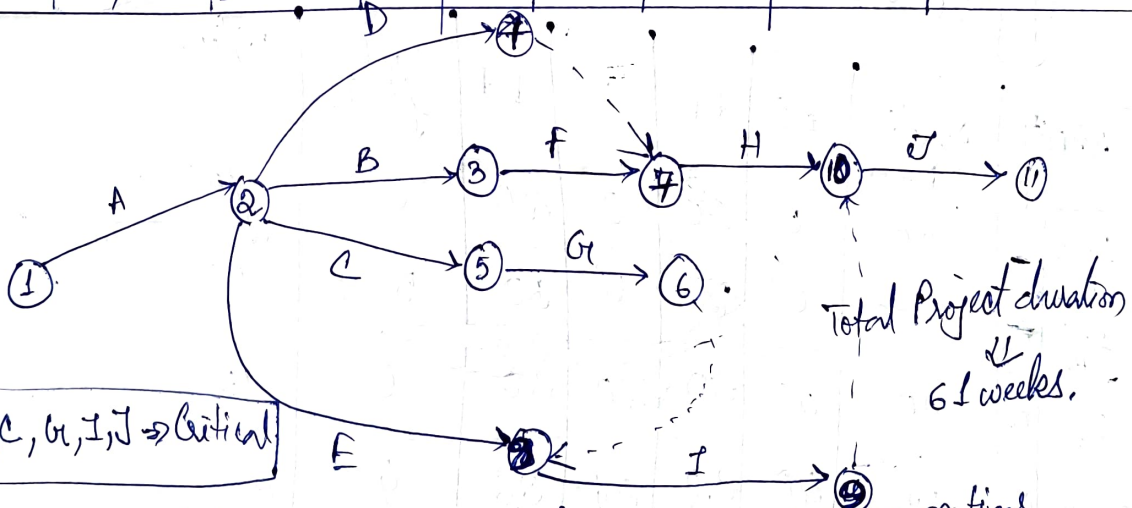
$0a + 1b \geq 2 \Rightarrow b \geq 2$



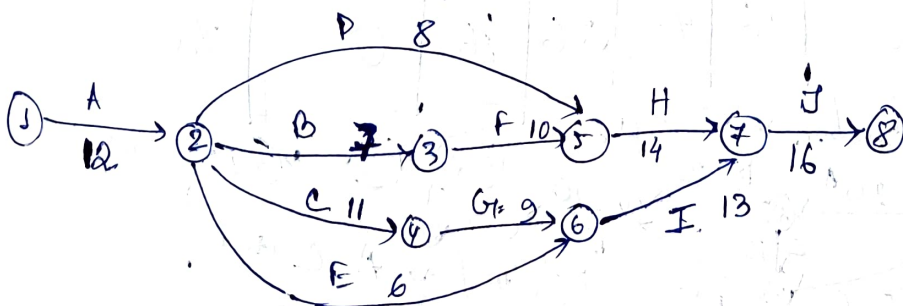
at $(4, 2) \Rightarrow z = 12 + 10 = 22$ (minimum)

Determine earliest and latest times, the total float for each activity, the critical activities and the project completion time.

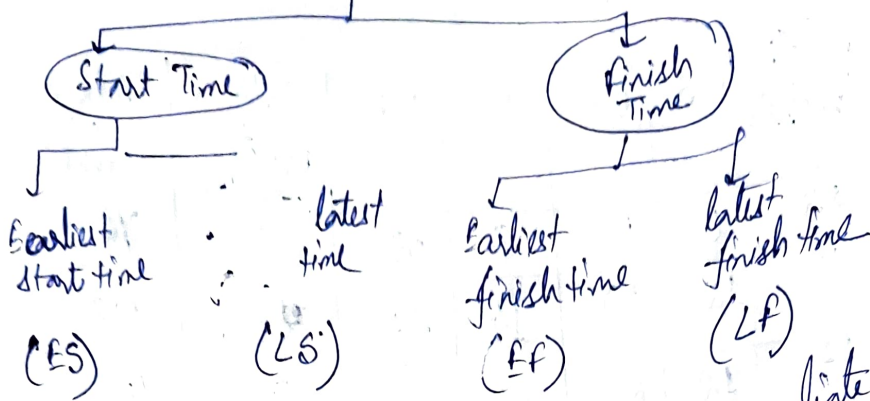
Activity	Immediate Predecessor Activity	Duration (weeks)	ES	EF	LS	LF	Total float	Critical or not
A	-	12	0	12	0	12	0	Yes
B	A	7	12	19	14	21	2	No
C	A	11	12	23	12	23	0	Yes
D	A	8	12	20	23	31	11	No
E	A	6	12	18	26	32	14	No
F	B	10	19	29	21	31	2	No
G	C	9	23	32	23	32	0	Yes
H	D, F	14	29	43	31	45	2	No
I	E, G	13	32	45	32	45	0	Yes
J	H, I	16	45	61	45	61	0	Yes



Dummy Activity -> Imaginary activity -> Consumes no resource or time i.e. duration zero -> to satisfy predecessor relationship



Activity Time



ES of an activity = Maximum EF of its immediate predecessors

$$EF = ES + \text{duration}$$

LF of an activity = Minimum LS of its successors

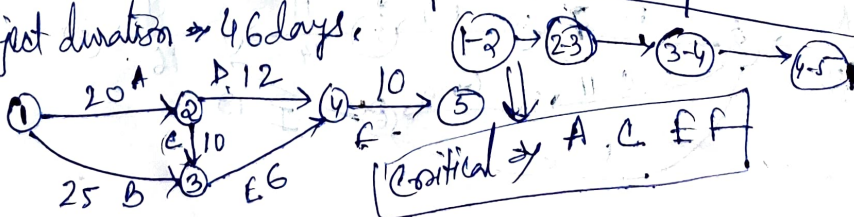
$$LS = LF - \text{duration}$$

$$\text{Float} = LS - ES = LF - EF$$

If float = 0, the activity is critical

Activity	P	Duration Days	ES	EF	LS	LF	Total float	Critical
1-2	-	20	0	20	0	20	0	Yes
1-3	A	25	0	25	5	30	5	No
2-3	A	10	20	30	20	30	0	Yes
2-4	A	12	20	32	24	36	4	No
3-4	B, C	6	30	36	30	36	0	Yes
4-5	D, E	10	36	46	36	46	0	Yes

Total Project duration = 46 days.



* Each day $\rightarrow$ 50 units of A, 25 units of B, 30 units of C produced.
 Total 100 man hours.
 Daily commitment for producing 20 units of A, 15 units of B & C.

Maximize! $12x_1 + 20x_2 + 45x_3$

Component availability: $x_1 \leq 50, y \leq 25, z \leq 30$

Assembly time: $0.8x_1 + 1.7x_2 + 2.5x_3 \leq 100$

Order commitments: $x \geq 20, y \geq 15$

Nonnegativity: $x, y, z \geq 0$

* Profits $\begin{matrix} \nearrow P_1 & 30 \\ \searrow P_2 & 40 \end{matrix}$

Maximize, $Z = 30x_1 + 40x_2$

Machine 1: $3x_1 + 2x_2 \leq 600$

Machine 2: $3x_1 + 5x_2 \leq 800$

Machine 3: $5x_1 + 6x_2 \leq 1100$

Non negativity constraints + $x_1 \geq 0, x_2 \geq 0$

	Iso-profit lines (Maximization problem)	Iso-cost lines (Minimization problem)
Approach	Begin with an arbitrary profit line, then move it parallel to itself in the direction of increasing profit until it reaches the furthest feasible point (away from origin)	Begin with an arbitrary cost line, then move it parallel to itself in the direction of decreasing cost until it reaches the nearest feasible point (towards origin)
Direction of Movement	The iso-profit line moves away from the origin towards higher objective function values	The iso-cost line moves towards the origin towards lower objective function values
Stopping point	The last point of contact with the feasible region represents optimal solution	The first point of contact with the feasible region represents the optimal sol ⁿ
Reason	Object is maximizing profit, we seek highest value of objective function that still intersects with feasible region	Objective is to minimize cost, we seek lowest possible value of the objective function that still intersects with feasible region

Capital Budgeting:

Investment decisions are generally called capital budgeting decisions.

Criteria:

- Discounted Cash Flow
- (i) NPV (Net Present Value) $\rightarrow$ What is the value of the investment
 - (ii) IRR $\rightarrow$ % of Return on Investment
 - (iii) Profitability Index $\rightarrow$ Profit ratio for every dollar spent
 - (iv) Payback period $\rightarrow$ Time to recover your investments.
- (Non-discounted cash flow)

Time value of Money $\rightarrow$ Future value of Money (FV)
 $\rightarrow$ Present value of Money (PV)

$n = \text{no. of years}$

$$PV = \frac{FV}{(1+K)^n}$$

$$FV = PV(1+K)^n$$

$K = \text{Discounted Rate in \%}$

NPV = Present value of all cash inflows - Present value of all cash outflows

If NPV > 0 (Accept the project)

NPV < 0 (Reject the project)

NPV $= 0$ (Accept considering other benefits)

If IRR $>$ Discount rate [Accept] NPV > 0

IRR $<$ Discount rate [Reject] NPV < 0

IRR = Discount Rate [May accept] NPV $= 0$

Discount Rate $\uparrow$
NPV $\downarrow$

Profitability Index

PI $> 1 \Rightarrow$ Accept

PI $< 1 \Rightarrow$ Reject

$$PI = \frac{\text{Cash Inflow}}{\text{Cash outflow}}$$

Payback Period/Break even point

Year	0	1	2	3
Investment	1,50,000			
Net Cashflow including depreciation		40,000	50,000	60,000
Depreciation		20,000	20,000	20,000

Tax Rate is 40%.

Net Cashflow including depreciation = sum of profit after tax and depreciation.

For Year 1, Profit ^{before} after tax (PBT) = $40,000 - 20,000 = 20,000$

$$PAT = PBT \times (1 - \text{Tax Rate})$$

$$\Rightarrow 20,000 = PBT \times (1 - 0.4)$$

$$\Rightarrow PBT = \frac{20,000}{0.6} = 33,333.33$$

$$\text{Tax} = PBT \times 0.4 = 13,333.33$$

$$\begin{aligned} \text{Cash Inflow} &= PBT - \text{Tax} + \text{Depreciation} \\ &= 33,333.33 - 13,333.33 + 20,000 \\ &= 40,000 \end{aligned}$$

For Year 2,
Cash Inflow = 50,000

For Year 3, Cash Inflow = 60,000

Year	Earning before tax (EBT)	Tax = EBT × Tax Rate	PAT (Profit after tax)
1	20,000	8,000	12,000
2	30,000	12,000	18,000
3	40,000	16,000	24,000

(PAT + Depreciation)
CFAT
32,000
38,000
44,000

Year	1	2	3	4
Cashflow	75,000	1,20,000	1,20,000	1,50,000

Initial Outlay = 3,00,000

Cost of capital = 10%

$$PV = \frac{FV}{(1+k)^n}$$

Year	Percentage 10%
1	0.909
2	0.826
3	0.751
4	0.683

Year	1	2	3	4
PV	68,175	99,120	90,120	1,02,450

Total PV of Inflows = 3,59,865

$$PI = \frac{3,59,865}{3,00,000} = 1.20$$

* Total PV of inflows @ 10% = 28.084 lakhs
NPV @ 10% = 28.084 - 25 = Rs 3.084 lakhs

Total PV of inflows @ 14% = Rs 24.863 lakhs

NPV @ 14% = 24.863 - 25 = Rs -0.137 lakhs.

$$IRR = r_1 + \left(\frac{NPV_1}{NPV_1 - NPV_2} \right) \times (r_2 - r_1)$$

$$= 10\% + \left(\frac{3.084}{3.084 - (-0.137)} \right) \times 4\%$$

$$= 10\% + (0.957) \times 4\%$$

$$= 13.83\%$$

costing 1,20,000, economic life $\rightarrow$ 5 years,
 salvage value $\rightarrow$ 20,000

Year 1 2 3 4 5

Output (units) 6000 6000 7,000 8,000 6,000

Selling price per unit $\rightarrow$ 10 Rs

Variable cost to sales ratio $\rightarrow$ 40%,

Operating fixed cost $\rightarrow$ 6,000 p.a.

Cost of capital $\rightarrow$ 10%,

corporate tax rate 50%,

$$\text{Depreciation} = \frac{1,20,000 - 20,000}{5} = 20,000 \text{ p.a.}$$

Yearly Cash flows $\rightarrow$

$$\text{Sales} = \text{Price} \times \text{Units}, \text{ VC} = 40\% \times \text{Sales}, \text{ FC} = 6000,$$

$$\text{EBIT} = \text{Sales} - \text{VC} = \text{FC} - \text{Depreciation}$$

$$\text{Net Income} = \text{EBIT} \times (1 - 0.5)$$

$$\text{CF} = \text{Net Income} + \text{Depreciation}$$

Year	Sales	V.C.	FC	EBIT	Net Income	aFAT
1	60,000	24,000	6000	10000	5000	25000
2	60,000	24,000	6000	10000	5000	25000
3	70,000	28,000	6000	16000	8000	28000
4	80,000	32,000	6000	22000	11000	31,000
5	60,000	24,000	6000	10000	5000	25,000

(25,000 + 20,000) $\rightarrow$ Salvage value
 $= 45,000$

Total PV cashflow @ 10% $\rightarrow$ 1,13,521

$$\text{NPV} = 1,13,521 - 1,20,000$$

$$= -6,479$$

$$\text{NPV} < 0$$

Idle $\rightarrow 15\%$ of the day $\rightarrow 72$ minutes

Wakes $\rightarrow 480$ minutes with an avg performance index 85

Drilled 400 items

Total time = 480 minutes

Working time = 408 minutes

Allowances 15% of total time = 72 minutes.

$$\begin{aligned}\text{Normal Time} &= \text{Working Time} \times \text{Performance Rating} \\ &= 408 \text{ minutes} \times 1.10 = 448.8 \text{ minutes}\end{aligned}$$

~~$$\begin{aligned}\text{Standard time} &= \text{Normal time} + \text{allowances} \\ &= 448.8 + 72 = 520.8 \text{ minutes}\end{aligned}$$~~

$$\text{Standard time per item} = \frac{528}{400} = 1.32 \text{ minutes/item}$$

$$\begin{aligned}\text{Standard time} &= \text{Normal time} \times \frac{100}{100 - \text{allowances}} \\ &= 448.8 \times \frac{100}{85} = 528\end{aligned}$$

$$1 \times 1.20 = 1.20$$

$$1.20 \times \frac{100}{100 - 10} = \frac{1.20 \times 100}{90} = \frac{12.0}{9}$$

$$\text{Allowances} = \text{Normal time} \times$$

$$\text{Allowances} = 448.8 \times 15\% =$$

~~$$10 \times 1.10 = 11.0$$~~

~~$$6 + 4.4$$~~

$$(10.4)$$

Productivity:-

Productivity is the ratio between output and input. It is quantitative relationship between what we produce and what we have spent to produce.

$$\text{Productive} = \frac{\text{Output}}{\text{Input}}$$

It is a state of mind-set. It is an attitude that seeks the continuous improvement of what exist. It is a conviction that one can do better today than yesterday, and that tomorrow will be better than today. It is the driving force or dynamism behind developing and upgrading the quality of industrial activities.

Importance + ① Productivity increases output.

- ② Emphasizes the efficient utilization of all factors of production.
- ③ Higher level of profit possible.
- ④ Attempts to eliminate wastage.
- ⑤ Chance of expansion, new business ventures and community support.
- ⑥ May pass on the benefits of lower costs to consumers in the form of lower prices.

Productivity is the measure of effectiveness i.e. doing the right thing efficiently.

Expressions of productivity

- Partial-factor prod.
- Multi factor prod.
- Total factor prod.

Techniques for measurement of Productivity

Aggregate Basis

Total Productivity

Output is compared with all inputs taken together
Hence, Total productivity Index = $\frac{\text{Total Output}}{\text{Total Input}}$
Total Output = Total Production of goods and services
Total Input = Labor + Material + Capital + Energy

Individual Basis

Partial Productivity factor

Output is compared with any one of the input factors.

Partial Productivity

Labor Productivity

$$= \frac{\text{Total output}}{\text{Labour Input}}$$

Material Productivity

$$= \frac{\text{Total Output}}{\text{Material Input}}$$

$$= \frac{\text{No. of units Produced}}{\text{Total Material cost}}$$

Machine Productivity

$$= \frac{\text{output in standard hrs}}{\text{Actual Machine hrs}}$$

Capital Productivity

$$= \frac{\text{Total output}}{\text{Capital employed}}$$

Factors affecting Productivity →

- (i) Capital investments in technology and equipment
- (ii) Economies of scale
- (iii) Work methods & procedures
- (iv) Systems
- (v) Technological changes
- (vi) Quality of management
- (vii) Quality & reliability of suppliers.

Efficiency = $\frac{\text{Actual output}}{\text{Effective Capacity}}$

Utilization = $\frac{\text{Actual output}}{\text{Design capacity}}$

(Measured as percentage)

$$N_r = \sum P_i D_i / T$$

N_r = no. of required machines

P_i = standard processing time for product i

D_i = demand for product i during the planning

T = processing time available ~ ~ ~

Cost-Volume Analysis

FC = Fixed Cost

VC = Variable Cost

$TC = VC + FC = \text{Total Cost}$

Total Revenue (TR) = Revenue per unit $\times Q$

Break-even point (BEP)

The volume of output at which total cost and total revenue are equal,

$$\text{Profit (P)} = TR - TC = (R \times Q) - (FC + V \times Q) \\ = Q(R - V) - FC$$

$$Q \times (\text{BEP}) = \frac{FC}{(R - V)} \Rightarrow \text{BEP} = \frac{\text{Fixed Cost}}{\text{Contribution}} = \frac{\text{Selling Price per unit} - \text{Variable cost per unit}}$$

$$\text{BEP (in terms of money value)} = \frac{FC}{R - V} \times R = \frac{FC}{\text{P/V Ratio}} \quad (\text{Profit volume})$$

$$\text{P/V Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$$

$$\text{P/V Ratio} = \frac{R - V}{R} \times 100$$

$$\text{Margin of Safety} = \text{Actual Sales} - \text{Break Even Sales} \text{ or } \frac{\text{Profit}}{\text{P/V Ratio}}$$

$$\text{M.S. Ratio} = \frac{\text{Margin of Safety}}{\text{Sales}} \times 100$$

Sales to get a desired profit, not just break even —

$$\text{Sales} = \frac{FC + \text{Desired Output (In Rs)}}{\text{P/V Ratio}}$$

$$(R - V) \text{ (Contribution)} = F.C. + \text{Profit}$$

$$\text{Depreciation (d)} = \frac{\text{Initial cost} - \text{Scrap value}}{\text{Service life}}$$

Plant layout - The success of an organization is also based on its plant layout. The systematic arrangement of men, machinery and equipment in a given plant area, so as to achieve maximum production with minimum cost is called plant layout. It means keeping the machinery and equipment within a given plant area. It can be defined as the process of determining a spatial location for the creation of physical production facilities suitable to manufacture a product.

Since, a layout once made cannot be changed/modified easily and without incurring considerable cost on one hand and disrupting the operations on the other hand, layout decisions are strategic decisions.

Various types of plant layout is based on the no. of products (P) and production quantity (Q) —

- ① Product layout (line layout)
- ② Process layout (Functional layout)
- ③ Combination layout
- ④ Fixed position layout.

① In this type of layout, all the machines are arranged in the sequence, as required to produce a specific product. It is called line layout as machines are arranged in a straight line. The raw materials are fed at one end and taken out as finished product to the other end. Special purpose machines are used which perform the required jobs quickly and reliably.

Advantages -

- (i) Faster and greater production
- (ii) Lower cost of material handling
- (iii) Product completes in lesser time
- (iv) Team work benefits
- (v) Easy monitoring
- (vi) Smooth and continuous work-flow
- (vii) Better coordination

Disadvantages -

- (i) Little flexibility
- (ii) Discontinuity in production lightly
- (iii) Monitoring each worker made difficult.
- (iv) High capital outlays
- (v) Threat of duplication.

② Process layout is also called functional layout. If the layout is designed for different products and for different processes, it is said to be called process layout. In this case, the machinery and equipment is arranged in a separate sections, for different products and for different operations, for producing two products different types of machinery and process to be followed.

Advantages +

- (i) Maximum utilization of resources

- (ii) Flexibility
- (iii) Continuity
- (iv) Interesting to workers
- (v) Easy monitoring
- (vi) Convenient to workers.
- (vii) Lower investment due to general purpose machines.

Disadvantages +

- (i) Higher wageable
- (ii) Higher material handling cost.
- (iii) Larger production cycle
- (iv) Monitoring may be complex
- (v) Higher inspection cost.
- (vi) Production planning and control is difficult
- (vii) More space requirement.

(2) In practice, plants are rarely laid out either in product or process layout form. Generally a combination of the two basic layouts is employed to derive the advantages of both systems of layout. (Ex: Refrigeration manufacturing) Process layout is used to produce various operations like stamping, welding, heat treatment being carried out in different work centers as per requirement. The final assembly of the product is done in product type layout.

(3) Fixed position layout is also called stationary layout. If the layout is designed in a fixed position it is said to be fixed layout. Here the material components remains stationary. Men, machinery tools etc. are brought to that location and assembled the product. Observed in large projects.

Advantages +

- (i) does not involve large investment

- (ii) High degree of flexibility

- (iii) Workers find it very interesting

Disadvantages +

- (i) Material handling cost will be very high.

- (ii) At times the resources may be under-utilized.

HRM + HRM is the study of activities regarding people working in an organization. It is a managerial function that tries to match an organization's needs to the skills and abilities of its employees.

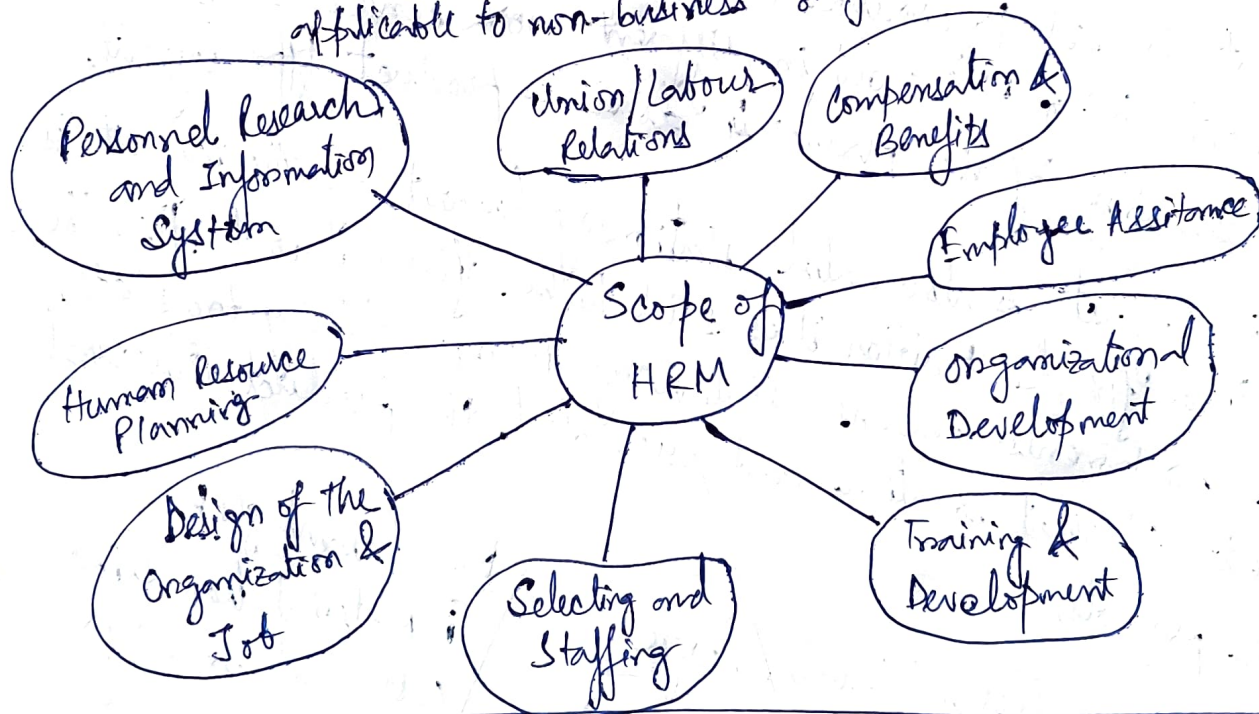
HRM is a management function concerned with hiring, motivating and maintaining people in an organization. It focuses on people in organizations, HRM is designing management systems to ensure that human talent is used effectively and efficiently to accomplish organizational goals.

HRM is the personnel funcⁿ which is concerned with procurement, development, compensation, integration and maintenance of the personnel of an organization for the purpose of contributing towards the accomplishments of the organization's objectives. Therefore, personnel management is the planning, organizing, directing and controlling of the performance of those operative functions.

HRM is the management funcⁿ that help managers to recruit, select, train and develop members for an organization. HRM is concerned with people's dimension in organizations.

Core →

- (i) It involves the application of management functions and principles.
- (ii) Decision relating to employees must be integrated.
- (iii) Decisions made influence the effectiveness of an organization.
- (iv) HRM functions are not confined to business establishments only but applicable to non-business organizations as well.



HRM objectives

Supporting functions

1. Societal Objectives

Legal Compliance
Benefits
Union-management relations

2. Organizational Objectives

Human Resource Planning
Employee Relations
Selection
Training & Development
Appraisal
Placement
Assessment

3. Functional Objectives

Appraisal
Placement
Assessment

4. Personal Objectives

Training and development
Appraisal
Placement
Compensation
Assessment

Work Measurement

$$\text{Standard Time} = \text{Set up Time} + \text{Time per piece} \times \text{No. of pieces produced}$$

$$\text{Standard Time} = \text{Normal Time} + \text{Allowance Value/Factor}$$

$$\text{Standard Time} = \text{Normal Time} \times \frac{100}{100 - \text{Allowance Rate}}$$

$$\text{Normal Time} = \text{Observed Time} \times \text{Rating factor}$$

$$\text{Performance Rating/Efficiency} = (\text{Observed Performance} / \text{Normal Performance}) \times 100$$

$$\text{Total Allowances (Station time)} = \text{Total Maintenance time} + \text{Interruption Time} + \text{Delay Time} + \text{Personal Time}$$

$$\text{Total Available Cycle Time} = \text{Total work period} - \text{Total allowances}$$

$$\text{Productive Hours} = \text{Time Available} / \text{No. of Hours}$$

PERT/CPM

$$\text{Earliest Start Time (ES)} = \text{Maximum EF of its immediate predecessors}$$

$$\text{EF} = \text{ES} + \text{duration}$$

$$\text{Latest Finish Time (LF)} = \text{Minimum LS of its successors}$$

$$\text{LS} = \text{LF} - \text{duration}$$

$$\text{Float} = \text{LS} - \text{ES} = \text{LF} - \text{EF}$$

If float = 0, the activity is critical.

Capital Budgeting

$$\text{EBIT (Earnings Before Interest \& Taxes)} = \text{Sales} - \text{Variable Cost} - \text{Fixed Cost} - \text{Depreciation}$$

$$\text{Tax} = \text{EBIT} \times \text{p\%}$$

$$\text{EAT} = \text{EBIT} - \text{Tax}$$

$$\text{Cash flow after Tax (CFAT)} = \text{EAT} + \text{Depreciation}$$

$$\text{NPV} = \text{Total PV of Cash Inflow} - \text{Initial Outlay}$$

$$\text{IRR} = r_1 + \left(\frac{\text{NPV}_1}{\text{NPV}_1 - \text{NPV}_2} \right) \times (r_2 - r_1)$$

$$\text{PI} = \frac{\text{Total PV of inflows}}{\text{Initial Outlay}} \quad (r_1 < r_2)$$

$$\text{PV} = \frac{\text{FV}}{(1+k)^n}$$

$$\text{Depreciation} = \frac{\text{Initial Cost} - \text{Scrap Value}}{\text{Service Life}}$$

UNIT: II- CAPITAL BUDGETING

SHORT ANSWERS:

Q.1. What are the characteristics of Capital Budgeting?

The capital budgeting decisions are important, crucial and critical business decisions due to following reasons:

- **Substantial expenditure:** Capital budgeting decisions involves the investment of substantial amount of funds. It is therefore necessary for a firm to make such decisions after a thoughtful consideration so as to result in the profitable use of its scarce resources.
- **Long time period:** The capital budgeting decision has its effect over a long period of time. These decisions not only affect the future benefits and costs of the firm but also influence the rate and direction of growth of the firm.
- **Irreversibility:** Most of the investment decisions are irreversible. Once they are taken, the firm may not be in a position to reverse them back.
- **Complex decision:** The capital investment decision involves an assessment of future events, which in fact is difficult to predict. Further it is quite difficult to estimate in quantitative terms all the benefits or the costs relating to a particular investment decision.

Q.2. Briefly explain the Capital Budgeting Process.

The extent to which the capital budgeting process needs to be formalised and systematic procedures established depends on the size of the organisation; number of projects to be considered; direct financial benefit of each project considered by itself; the composition of the firm's existing assets and management's desire to change that composition; timing of expenditures associated with the projects that are finally accepted.

(i) Planning (ii) Evaluation (iii) Selection (iv) Implementation (v) Control (vi) Review

Q.3. Write short notes on the kinds of capital budgeting decisions.

Generally capital investment decisions are classified in two ways:

A. On the basis of firm's existence: The capital budgeting decisions are taken by both newly incorporated firms as well as by existing firms. These decisions may be classified as follows:

(i) Replacement and Modernisation decisions: The replacement and modernisation decisions aim at to improve operating efficiency and to reduce cost.

(ii) Expansion decisions: Existing successful firms may experience growth in demand of their product line, they may consider proposal to add capacity to existing product line.

(iii) Diversification decisions: These decisions require evaluation of proposals to diversify into new product lines, new markets etc. for reducing the risk of failure by dealing in different products or by operating in several markets.

B. On the basis of decision situation: The capital budgeting decisions on the basis of decision situation are classified as follows:

(i) Mutually exclusive decisions: The decisions are said to be mutually exclusive if two or more alternative proposals are such that the acceptance of one proposal will exclude the acceptance of the other alternative proposals.

(ii) Accept-reject decisions: The accept-reject decisions occur when proposals are independent and do not compete with each other. The firm may accept or reject a proposal on the basis of a minimum return on the required investment.

(iii) Contingent decisions: The contingent decisions are dependable proposals. The investment in one proposal requires investment in one or more other proposals.

Q.4. Name the various methods of capital budgeting.

In order to maximise the return to the shareholders of a company, it is important that the best or most profitable investment projects are selected. Because the results for making a bad long-term investment decision can be both financially and strategically devastating, particular care needs to be taken with investment project selection and evaluation. There are a number of techniques available for appraisal of investment proposals and can be classified below:

A. Traditional or Non-Discounting methods:

- (i) Payback Period Method.
- (ii) Accounting Rate of Return Method.

B. Time Adjusted or Discounted Cash Flows Methods:

- (i) Net Present Value Method.
- (ii) Internal Rate of Return Method.
- (iii) Profitability Index Method.

Q.5. Write a short note on Pay-back period.

Payback period is one of the non-discounted cash flow methods of capital budgeting. Pay-back period is the time required to recover the initial investment in a project.

$$\text{Pay - back period} = \frac{\text{Initial investment}}{\text{Annual cash inflows}}$$

Merits of Pay-back method

- It is easy to calculate and simple to understand.
- Pay-back method provides further improvement over the accounting rate return.
- Pay-back method reduces the possibility of loss on account of obsolescence.

Demerits of Pay-back method

- It ignores the time value of money.
- It ignores all cash inflows after the pay-back period.
- It is one of the misleading evaluations of capital budgeting.

Q.6. What is Accounting Rate of Return.

Accounting rate of return means the average rate of return or profit taken for considering the project evaluation. This method is one of the traditional methods for evaluating the project proposals.

$$\text{Accounting rate of return} = \frac{\text{Average annual net income}}{\text{Investment}}$$

Merits of Accounting Rate of Return method:

- It is easy to calculate and simple to understand.
- It is based on the accounting information rather than cash inflow.
- It is not based on the time value of money.
- It considers the total benefits associated with the project.

Demerits of Accounting Rate of Return method:

- It ignores the time value of money.
- It ignores the reinvestment potential of a project.
- Different methods are used for accounting profit.

Accept/Reject criteria

If the actual accounting rate of return is more than the predetermined required rate of return, the project would be accepted. If not, it would be rejected.

Q.7. Explain Net Present Value.

NET PRESENT VALUE METHOD

The net present value method is a modern method of evaluating investment proposals. This method takes into consideration the time value of money.

The present values of all inflows and outflows of cash occurring during the entire life of the project is determined separately for each year by discounting these flows by the firm's cost of capital or a pre-determined rate.

"NET PRESENT VALUE IS THE DIFFERENCE BETWEEN THE TOTAL PRESENT VALUE OF FUTURE CASH INFLOWS AND THE TOTAL PRESENT VALUE OF FUTURE CASH OUTFLOWS."

Merits of Net Present Value Method:

- It recognizes the time value of money.
- It considers the total benefits arising out of the proposal.
- It is the best method for the selection of mutually exclusive projects.
- It helps to achieve the maximization of shareholders' wealth.

Demerits of Net Present Value Method:

- It is difficult to understand and calculate.
- It needs the discount factors for calculation of present values.
- It is not suitable for the projects having different effective lives.

Q.8. What is Profitability Index?

PROFITABILITY INDEX METHOD OR BENEFIT COST RATIO

It is a time -adjusted method of evaluating the investment proposals. Profitability index also called as Benefit Cost Ratio (B/C) or 'Desirability factor' is the relationship between present value of cash inflows and the present value of cash outflows.

$$\text{Profitability Index} = \frac{\text{Present Value of Cash Inflows}}{\text{Present Value of Cash Outflows}}$$

The proposal is accepted if the profitability index is more than one and is rejected in case the profitability index is less than one. The profitability index method is most suitable, particularly when the costs of the projects differ significantly.

Essay Questions:

Q.1. What do you understand by Capital Budgeting Process? Enumerate briefly the major steps involved in capital budgeting.

CAPITAL BUDGETING PROCESS

The extent to which the capital budgeting process needs to be formalised and systematic procedures established depends on the size of the organisation; number of projects to be considered; direct financial benefit of each project considered by itself; the composition of the firm's existing assets and management's desire to change that composition; timing of expenditures associated with the projects that are finally accepted.

(i) Planning: The capital budgeting process begins with the identification of potential investment opportunities. Opportunities having little merit are rejected and promising opportunities are advanced in the form of a proposal to enter the evaluation phase.

(ii) Evaluation: This phase involves the determination of proposal and its investments, inflows and outflows. Investment appraisal techniques, ranging from the simple payback method and accounting rate of return to the more sophisticated discounted cash flow techniques, are used to appraise the proposals. The technique selected should be the one that enables the manager to make the best decision in the light of prevailing circumstances.

(iii) Selection: Considering the returns and risks associated with the individual projects as well as the cost of capital to the organisation, the organisation will choose among projects so as to maximise shareholders' wealth.

(iv) Implementation: When the final selection has been made, the firm must acquire the necessary funds, purchase the assets, and begin the implementation of the project.

(v) Control: The progress of the project is monitored with the aid of feedback reports. These reports will include capital expenditure progress reports, performance reports comparing actual performance against plans set and post completion audits.

(vi) Review: When a project terminates, or even before, the organisation should review the entire project to explain its success or failure. This phase, may have implication for firms planning and evaluation procedures. Further, the review may produce ideas for new proposals to be undertaken in the future.

Q.2. Explain briefly the following methods of capital budgeting bringing out the advantages and disadvantages of each: (a) Pay-back period method (b) Accounting Rate of Return method.

In order to maximise the return to the shareholders of a company, it is important that the best or most profitable investment projects are selected. There are a number of techniques available for appraisal of investment proposals and can be classified below:

I. Traditional or Non-Discounting methods:

- (i) Payback Period Method.
- (ii) Accounting Rate of Return Method.

II. Time Adjusted or Discounted Cash Flows Methods:

- (i) Net Present Value Method.
- (ii) Internal Rate of Return Method.
- (iii) Profitability Index Method.

❖ PAYBACK PERIOD METHOD:

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❖ ACCOUNTING RATE OF RETURN METHOD:

Accounting rate of return means the average rate of return or profit taken for considering the project evaluation. This method is one of the traditional methods for evaluating the project proposals.

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- It ignores the reinvestment potential of a project.
- Different methods are used for accounting profit.

Accept/Reject criteria

If the actual accounting rate of return is more than the predetermined required rate of return, the project would be accepted. If not it would be rejected.

Q.3. "NPV method ensures maximization of wealth of the shareholders in the long run". Comment.

NET PRESENT VALUE METHOD

The net present value method is a modern method of evaluating investment proposals. This method takes into consideration the time value of money and attempts to calculate the return on investments by introducing the factor of time element. It recognises the fact that a rupee earned today is worth more than the same rupee earned tomorrow. The present values of all inflows and outflows of cash occurring during the entire life of the project is determined separately for each year by discounting these flows by the firm's cost of capital or a pre-determined rate.

"NET PRESENT VALUE IS THE DIFFERENCE BETWEEN THE TOTAL PRESENT VALUE OF FUTURE CASH INFLOWS AND THE TOTAL PRESENT VALUE OF FUTURE CASH OUTFLOWS."

Merits of Net Present Value Method:

- It recognizes the time value of money.
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- It helps to achieve the maximization of shareholders' wealth.

Demerits of Net Present Value Method:

- It is difficult to understand and calculate.
- It needs the discount factors for calculation of present values.
- It is not suitable for the projects having different effective lives.

Q.4. "The virtue of IRR method is that it does not require the pre- calculation of the required rate of return", Critically examine.

INTERNAL RATE OF RETURN METHOD

The internal rate of return method is a modern technique of capital budgeting that takes into account the time value of money. It is also known as 'time adjusted rate of return', 'discounted cash flow', 'discounted rate of return,' 'yield method,' and 'trial and error yield method.

In the net present value method, the net present value is determined by discounting the future cash flows of a project at a predetermined or specified rate called the cut-off rate. But under the internal rate of return method, the cash flows of a project are discounted at a suitable rate by hit and trial method, which equates the net present value so calculated to the amount of the investment.

Under this method, since the discount rate is determined internally, this method is called as the internal rate of return method.

The internal rate of return can be defined as that rate of discount at which the present value of cash-inflows is equal to the present value of cash outflows.

Merits of Internal Rate of Return Method:

- It considers the time value of money.
- It takes into account the total cash inflow and outflow.
- It does not use the concept of the required rate of return.
- It gives the approximate/nearest rate of return.

Demerits of Internal Rate of Return Method:

- It involves complicated computational method.
- It produces multiple rates which may be confusing for taking decisions.
- It is assuming that all intermediate cash flows are reinvested at the internal rate of return.
- The results of NPV method and IRR method may differ when the projects under evaluation differ in their size, life and timings of cash flows.